


 Name: Answer Sheet

 Total Marks: ³⁶40 marks
 Total Time: 45 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Resources Allowed: 1 page A4 notes (1 side), calculator.

Question 1
11 marks

Calculate the mean, median, mode and range for the following set of data, once including the outlier and the other without.

~~23, 32, 16, 37, 15, 31, 2, 25,~~
~~2, 15, 16, 23, 25, 31, 32, 37~~ — 1 mark for ordering

	With outlier	Without outlier
Mean	2 marks. $\frac{181}{8} = 22.625$	$\frac{179}{7} = 25.57$
Median	$\frac{23+25}{2} = 24$	25
Mode	No Mode.	No Mode
Range	$37 - 2 = 35$	$37 - 15 = 22$

 a) Identify the outlier in the data: 2 — 1 mark

b) What effect does the outlier have on the data (What happens to the mean, median, mode and range)?

 Mean — ~~decreased~~ Increased ✓

Median — Increased ✓

Mode — No Change ✓

Range — Significantly Smaller / Reduced ✓

1 mark each

Question 2
3 marks

What type of data are the following?

Choose from categorical, discrete numerical or continuous numerical:

a) the number of hours spent playing video games each week

Discrete Numerical ✓

b) the favourite pizza topping of each of the students in a class

Categorical ✓

c) the length of time that a year 8 student can spend running

Continuous Numerical ✓

Question 3

3 marks ~~7 marks~~

Classify each of the following data collections according to whether they are a **sample** or **census**

- a) You want to know the favourite sport to watch in Perth, so you ask every student and Baldivis Secondary College

Sample ✓

- b) Number of hours spent completing homework each week by your class by surveying each student in your class.

Census ✓

- c) Asking every student that walks through the gates of Baldivis Secondary College how they travel to school.

Sample. ✓

Question 4

2 marks

Use only the numbered cards below to answer the following questions.



Choose 3 numbers from the cards above so that there is:

- a) no mode a range of 6 a mean of 6

3

6

9



1 mark

Choose 4 numbers from the cards above so that there is:

- b) a mean of 4 a mode of 4

4

4

~~3~~

~~5~~

3

5

or

1

7

~~then~~
Only one of these ~~ans~~ answers is required.

Question 5

10 marks

The times taken by 24 students to complete a mathematical puzzle are shown below in minutes:

5, 22, 8, 13, 7, 9, 6, 8, 12, 10, 4, 9, 3, 19, 18, 9, 14, 2, 5, 15, 21, 11, 5, 17

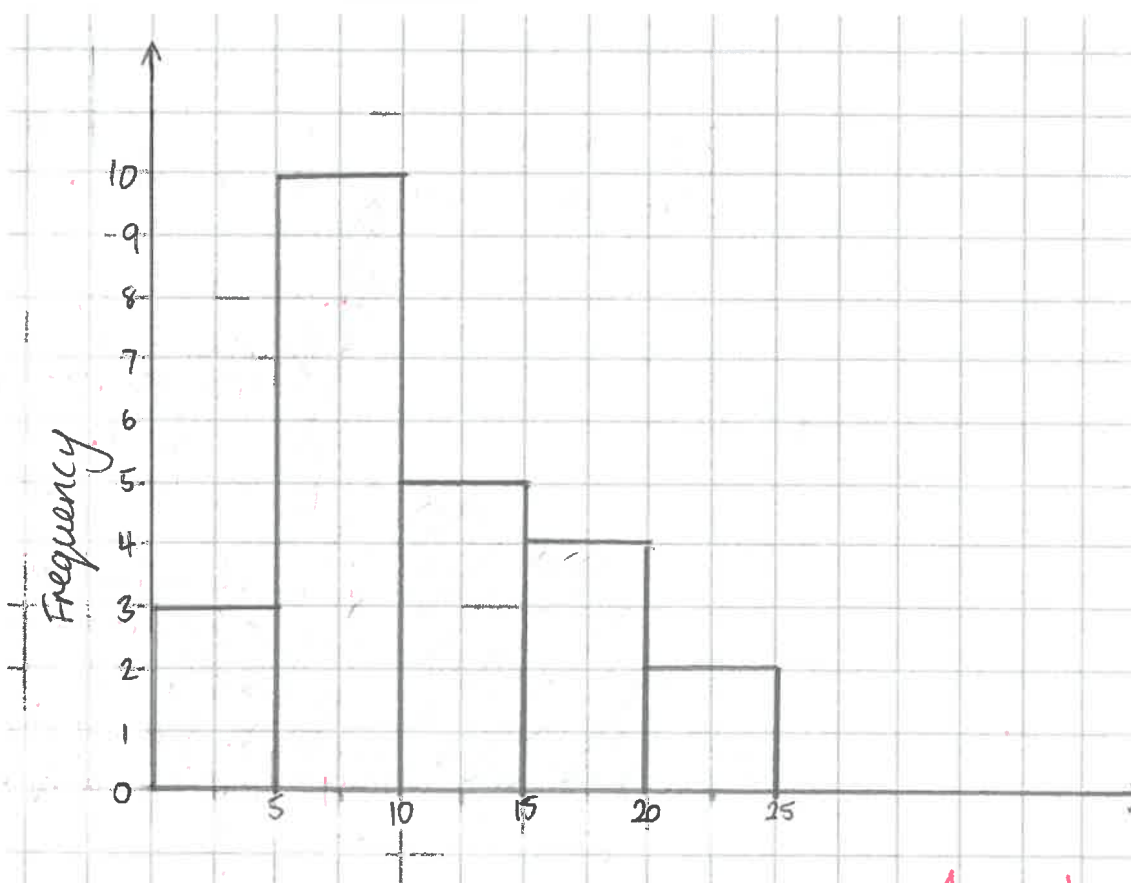
a) Create a grouped frequency table

Time taken (minutes)	Tally	Frequency
$0 \leq x < 5$		3
$5 \leq x < 10$		10
$10 \leq x < 15$		5
$15 \leq x < 20$		4
$20 \leq x < 25$		2

1 mark per
correct line.

b) Construct a histogram using the data above

Mathematical Puzzle Results ✓ 1 mark



Time Taken (minutes)

Allow for follow through
from part A.

1 mark - ruler used.

1 mark - labelled axes

1 mark - Appropriate
scales chosen
to represent data

1 mark - Columns equal
width and touching

Question 6

3 marks



a) Would you describe the shape as symmetrical or skewed?

Skewed ✓

b) How many workers were over the age of 60?

$$3 + 1 = 4 \quad \checkmark$$

c) How many workers were interviewed in total?

$$8 + 12 + 7 + 6 + 3 + 1 = 37 \quad \checkmark$$

Question 7

4 marks

Students were asked to find the mean, median, mode and range for the Stem and Leaf plot. Tom gave the following answers. **CIRCLE** which of his answers are **incorrect**? AND give the correct answer showing working out.

stem	leaf
2	1, 5, 6
3	0, 0, 1
4	0

Mean = 29

median = 26

mode = 30

range = 34

If only median and range are circled, award 1 mark.

1 mark

- Working out/mathematical justification for mean being correct.
No working out, no mark.

1 mark - justifying median either by crossing out or listing values & finding median or equivalent.
No working out, no mark.

$$\text{Mean} = \frac{21 + 25 + 26 + 30 + 30 + \cancel{30} + 40}{7} = 29$$

$$\text{Median} = 30 \quad (\text{From Table})$$

$$\text{Mode} = 30 \quad \checkmark$$

$$\text{Range} = 40 - 21 = 19 \quad \checkmark$$

Working out required.
No working out award ½ mark.

END OF TEST 😊